Bilkent University

Department Of Computer Engineering

Bilkent Information Office Web Page

CS-319 Section 1

Group 2

Deliverable-3 1st Iteration

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Top-2 Design Goals

The design goals stem from the analysis of non-functional requirements, which aim to make the software more viable. While designing the project, these two design goals will be prioritized, even if potential trade-offs occur, which will be noted in this report.

1. Usability

- The core design goal of the Information Office Web Page will be usability. The website should be easy to use since the web application is designed to introduce our university to high schools, prospective students, and their parents. We specified our target design to aim that users who use the app to obtain information about the university won't be confused while navigating our website. The main page design of the Information Office Web Page will cover all relevant details about the visiting appointment, seminars, FAQ, the university booklet, and terms and conditions. While obtaining all these options, we will use a table layout and oriented margins to each point so that it can easily be noticeable by the users. In addition, we will provide the relevant details immediately after that click, ensuring that the user can see all important topics that they search for within one or two clicks from the main page.
- Another point of the usability goal is that the director and Information Office members can easily adapt to the usage of our app. We will implement a dynamic website for them to prioritize their main responsibilities with ease. For instance, for each user type (Guide, advisor, head secretary, director), the dashboard will differ according to their main duties, including the to-do list that they can check and update after every login. The sections on the dashboard page will be clickable, allowing users to navigate the detailed page of the clicked entry (The tour section will redirect to the tour page). Also, the website contains categories to click on from the left side of the page, such as tours, guide schedules, high school analytics, and profile pages. Thus, the website lowers the possible confusion that Information Office members could experience in the first few days after starting to use the website.
- Lastly, managing the tour operations quickly will increase the performance of the information office team members since the most time-consuming part is managing the tours. To provide this, the website will rely on a notification system that performs by alerting all related users within ~1 second. This enhances usability by informing information office members of new updates about upcoming tours. To catch their attention, we will also provide a red dot on the notification icon to help them keep up with the new updates.

2. Reliability

- The second important design goal of the Information Office Web Page will be reliability. The website must consistently deliver accurate information about the tour operations and perform important actions without errors, such as assigning a guide to a tour or canceling an event. In order to achieve this, we will consider lots of corner cases tour operations may create and test them after our implementation.
- User data, tour schedules, and other critically sensitive information about the application will be stored securely and updated in real-time. Measures will be taken to prevent data loss or corruption during updates or system crashes. For instance, we will use PostgreSQL to ensure data integrity, support for ACID transactions, and real-time performance. PostgreSQL's key features, like point-in-time recovery and replication, can further enhance data safety and system availability.
- The system must accommodate approximately 100 concurrent users to guarantee seamless access. Spring boot has @Async annotation to enable asynchronous processing, which will be utilized. By running tasks asynchronously, the system can handle more requests concurrently without getting blocked by tasks.

3. Possible Trade-offs

3.1. Usability vs. Performance

Ensuring the website is highly usable will force us to use additional resources, such as loading visual elements for the detailed dashboards and structural layouts. While this improves user experience, it could slightly increase page load times since this heavy design can lead to higher memory consumption and slower startup times compared to lightweight frameworks.

3.2. Usability vs. Security

For the sake of usability, we will not be planning to implement a login page for prospective students and counselors. Although this eases user experience, our app can encounter some security issues, such as fake appliances for the high school tours by non-counselors or fake registration to the seminars.

3.3. Reliability vs Complexity

Enabling the system to handle approximately 100 concurrent users using asynchronous processing (@Async annotation in Spring Boot) brings some complexity to handle. We need to carefully design the order of execution since asynchronous tasks may not execute in a predictable order. This can be problematic for operations requiring strict sequencing, such as processing dependent tasks. As a result, we need a good design to overcome thread management issues for enabling the app for 100 concurrent users.

SubSystem Decomposition Diagram

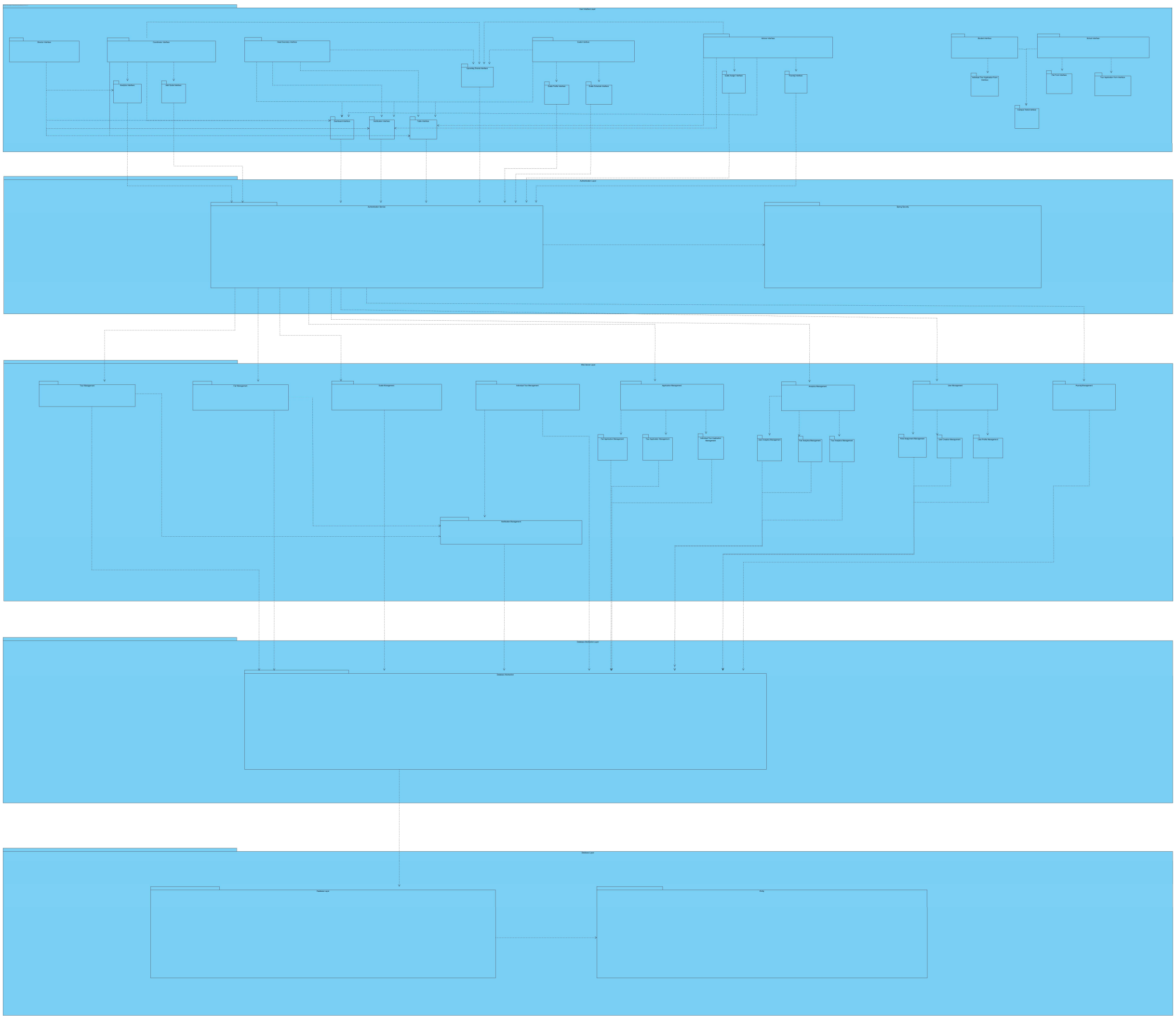

Figure 1. Subsystem Decomposition Diagram